APPLICATIONS OF DEEP LEARNING FOR NLP

The field of natural language processing is shifting from statistical methods to neural network methods. The followings are natural language processing tasks where deep learning methods are achieving some advancement.

1. Text Classification (text categorization)

The goal of text categorization is to classify the topic or theme of a document.

A popular classification example is sentiment analysis where class labels represent the emotional tone of the source text such as "*positive*" or "*negative*".

Below are a 3 more examples:

- Spam filtering, classifying email text as spam or not.
- Language identification, classifying the language of the source text.
- Genre classification, classifying the genre of a fictional story.

2. Language Modeling

The problem is to predict the next word given the previous words. The task is fundamental to speech or optical character recognition, and is also used for spelling correction, handwriting recognition, and statistical machine translation.

A language model learns the probabilistic relationship between words such that new sequences of words can be generated that are statistically consistent with the source text.

Alone, language models can be used for text or speech generation; for example:

- Generating new article headlines.
- Generating new sentences, paragraphs, or documents.
- Generating suggested continuation of a sentence.

3. Speech Recognition

Speech recognition is the problem of understanding what was said.

Given an utterance of text as audio data, the model must produce human readable text.

Some examples include:

- Transcribing a speech.
- Creating text captions for a movie or TV show.
- Issuing commands to the radio while driving.

4. Caption Generation

Caption generation is the problem of describing the contents of an image.

Given a digital image, such as a photo, generate a textual description of the contents of the image.

This is not just an application for blind, but also in generating human readable text for image and video data that can be searched, such as on the web.

5. Machine Translation

Machine translation is the problem of converting a source text in one language to another language.

6. Document Summarization

Document summarization is the task where a short description of a text document is created.

Some examples of document summarization include:

- Creating a heading for a document.
- Creating an abstract of a document.

7. Question Answering

Question answering is the problem where given a subject, such as a document of text, answer a specific question about the subject.

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